Q on Lect 7 : 11 Part 2 Made By : Ahmed Tamer

Lecture 7 — Intangible Assets other than Goodwill

Example 1 — Customer List (Green Market Inc.)

On January 1, 2021, Green Market Inc. acquires the customer list of a large newspaper for 4 years at $700,000. Green Market expects to benefit from the information over only 3 years, and it can sell the customer list after this period for $100,000. **Required:** Record the purchase on January 1, 2021, and the amortization at the end of each year.

الخطوة الأولى: تسجيل الشراء في 1/1/2021

الشركة اشترت customer list بـ $700,000 كاش. ببساطة بنسجل الـ asset بتاعنا وبنقابله بالكاش اللي خرج.

Accounts	Dr.	Cr.
Customer list	700,000	
Cash		700,000

الخطوة الثانية: تحديد فترة الاستهلاك

القاعدة بتقول دايماً خد الأقل بين الـ Useful Life والـ Legal Life.

- Legal Life = 4 سنين (ده اللي العقد بيقوله)
- Useful Life = 3 سنين (ده اللي الشركة قدّرت إنها هتستفيد فيه فعلاً)
- إذن بناخد الـ **3 سنين** لأنها الأقل

الخطوة الثالثة: حساب الـ Amortization

بنطبق الـ Straight-line method وهو الـ default للـ intangibles.

- Residual Value =
- $100,000 لأن الشركة ممكن تبيعه بعد 3 سنين بالسعر ده

Amortization Expense = (700,000 − 100,000) ÷ 3 = 600,000 ÷ 3 = $200,000 per year

القيد ده بيتكرر 3 مرات: في 31/12/2021 و31/12/2022 و31/12/2023

Accounts	Dr.	Cr.
Amortization expense — Customer list	200,000	
Customer list (or Accumulated amortization)		200,000

Example 2 — Franchise (National Corporation & Samsung)

Assume that National corporation acquired a franchise from Samsung Inc. at a cost of $400,000 on April 1, 2024. The agreement stated that the franchise grants National corporation the right to sell Samsung products for a period of 10 years, while National corporation estimated the useful life of the franchise to be only 8 years. **Required:** Record the acquisition on 1/4/2024 and the amortization at the end of 2024.

الخطوة الأولى: تسجيل الاستحواذ في 1/4/2024

Accounts	Dr.	Cr.
Franchise	400,000	
Cash		400,000

الخطوة الثانية: تحديد الفترة وحساب الـ Amortization

- Legal Life = 10 سنين (مدة العقد مع Samsung)
- Useful Life = 8 سنين (تقدير الشركة)
- بناخد الأقل = **8 سنين**

الـ franchise بدأ من 1 أبريل 2024، يعني من 1/4/2024 لـ 31/12/2024 = **9 شهور** بس مش سنة كاملة.

Annual Amortization = (400,000 − 0) ÷ 8 = $50,000 per year 2024 Amortization = 50,000 × 9/12 = $37,500

الـ Residual Value = zero لأن مفيش قيمة متبقية للـ franchise بعد ما تنتهي.

Accounts	Dr.	Cr.
Amortization expense — Franchise	37,500	
Franchise (or Accumulated amortization — Franchise)		37,500

Example 3 — Internally Created Patent (Al-Ahly Company)

On November 1, 2023, Al-Ahly company started an R&D project to develop a patent. During November and December, costs were $35,000 (research) and $45,000 (development). On March 1, 2024, the project became economically viable and the patent was registered for 10 years at $90,000. The company estimates the patent's useful life to be 12 years. **Required:** Record R&D costs during Nov and Dec 2023 — Amortization at end of 2023 if any — Registration cost on March 1, 2024 — Amortization at end of 2024 if any — Effect on financial statements.

الخطوة الأولى:

- المشروع اتبدأ 1/11/2023 وفيه مرحلتين Research وDevelopment
- نقطة الـ Economic Viability اتحددت في 1/3/2024
- القاعدة: كل التكاليف **قبل** الـ economic viability بتتسجل expense، وبعديها بتتكبل كـ asset

في نوفمبر وديسمبر 2023، الشركة أنفقت $35,000 + $45,000 = **$80,000** كلها جاية قبل الـ economic viability فكلها expense.

الخطوة الثانية: R&D Costs في نوفمبر وديسمبر 2023

Accounts	Dr.	Cr.
Research and development expenses	80,000	
Cash (35,000 + 45,000)		80,000

الخطوة الثالثة: Amortization في 31/12/2023

في 31/12/2023 مفيش intangible asset اتسجل لحد دلوقتي. الشركة لسه في مرحلة الـ R&D وما وصلتش للـ economic viability. **No entry required**.

الخطوة الرابعة: Registration Cost في 1/3/2024

في 1/3/2024 اتحققت الـ economic viability والـ patent اتسجل بـ $90,000. الـ registration cost بتتكبل لأن التسجيل دليل على إن الأصل وصل لمرحلة الاستخدام الاقتصادي.

Accounts	Dr.	Cr.
Patent	90,000	
Cash		90,000

الخطوة الخامسة: Amortization في 31/12/2024

- Legal Life = 10 سنين (مدة التسجيل القانوني)
- Useful Life = 12 سنين (تقدير الشركة)
- **بناخد الأقل = 10 سنين**

الـ patent بدأ في 1/3/2024، فمن 1/3/2024 لـ 31/12/2024 = 10 شهور

Annual Amortization = (90,000 − 0) ÷ 10 = $9,000 per year 2024 Amortization = 9,000 × 10/12 = $7,500

Accounts	Dr.	Cr.
Amortization expense — Patent	7,500	
Patent (or Accumulated amortization — Patent)		7,500

الخطوة السادسة: Effect on Financial Statements

Statement of Financial Position على 31/12/2024:

- Patent = Cost 90,000 − Accumulated Amortization 7,500 = **$82,500** (في قسم Intangible Assets)

Income Statement لسنة 2024:

- Amortization expense = $7,500

بالنسبة لـ 2023:

- Income Statement فيها R&D expenses $80,000 فقط
- مفيش effect على الـ Balance Sheet في 2023 لأن مفيش asset اتسجل

> Example 4 — Purchased Patent with Legal Defense (Sonic Company)

On January 1, 2010, Sonic company purchased a patent for $90,000 and registered it at fees of $10,000. Legal life = 23 years, useful life estimated = 20 years. On January 1, 2015, Sonic spent $51,000 to successfully prosecute a patent violation, and as a result the useful life was extended for additional 6 years. **Required:** Purchase entry on 1/1/2010 — Amortization at end of 2010 — Legal cost on 1/1/2015 — Amortization at end of 2015.

> الخطوة الأولى: تسجيل الشراء في 1/1/2010

الـ cost الكلية = Purchase Price + Registration Fees = 90,000 + 10,000 = **$100,000**

الـ Registration Fees بتضاف للـ cost لأنها ضرورية عشان الأصل يبقى جاهز للاستخدام.

Accounts	Dr.	Cr.
Patent	100,000	
Cash		100,000

> الخطوة الثانية: Amortization في 31/12/2010

- Legal Life = 23 سنة
- Useful Life = 20 سنة
- بناخد الأقل = **20 سنة**

Amortization = (100,000 − 0) ÷ 20 = $5,000 per year

Accounts	Dr.	Cr.
Amortization expense — Patent	5,000	
Patent (or Accumulated amortization)		5,000

الخطوة الثالثة: Legal Cost في 1/1/2015 — نجاح في الـ Defense

الشركة نجحت في الدفاع عن الـ patent إذن الـ legal cost بتتكبّل وبتضاف للـ patent balance.

حساب الـ carrying value قبل تسجيل القيد:

- Patent cost الأصلي = 100,000
- ناقص Amortization من 2010 لـ 2014 = 5 سنين × 5,000 = (25,000)
- زائد Legal cost الجديدة = 51,000
- **Patent balance على 1/1/2015 = 126,000$**

Accounts	Dr.	Cr.
Patent	51,000	
Cash		51,000

الخطوة الرابعة: Amortization في 31/12/2015

النجاح في الـ defense أدى لمد العمر بـ 6 سنين إضافية، لازم نعيد حساب الـ remaining life.

الـ Remaining Useful Life على 1/1/2015:

- Useful Life الأصلية 20 سنة ناقص 5 سنين انقضوا (2010-2014) = 15 سنة متبقية
- زائد 6 سنين إضافية من النجاح = **21 سنة**

الـ Remaining Legal Life على 1/1/2015:

- Legal Life الأصلية 23 سنة ناقص 5 سنين = **18 سنة متبقية**

بناخد الأقل بين الـ 21 سنة والـ 18 سنة = **18 سنة**

Amortization = 126,000 ÷ 18 = $7,000 per year

Accounts	Dr.	Cr.
Amortization expense — Patent	7,000	
Patent (or Accumulated amortization)		7,000

لو الشركة فشلت في الـ Defense على 1/1/2015:

الـ legal cost بتبقى expense والـ carrying value على 1/1/2015 = 100,000 – 25,000 = 75,000$ بتتسجل كاملة كـ loss:

Accounts	Dr.	Cr.
Legal expense	**51,000**	
Cash		**51,000**
Accounts	Dr.	Cr.
Patent loss	75,000	
Patent		75,000

Lecture 8 — Impairment of Intangible Assets other than Goodwill

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Example 1 — Broadcast License Indefinite-Life (Arcon Radio Co.)

On 1/1/2023, Arcon Radio Co. purchased a broadcast license for $2,000,000 and reports it as an indefinite-life intangible asset. At end of 2023, impairment test was performed.

Required: Journal entry for impairment (if any) in two cases:

- Case 1 — Fair value less costs to sell = $2,500,000
- Case 2 — Fair value less costs to sell = $1,500,000

النقاط الأساسية قبل الحل:

- الـ Broadcast License ده **indefinite-life** يعني مفيش amortization خالص
- الـ Carrying Value = Cost = $2,000,000 لأنه ما اتستهلكش
- للـ indefinite-life assets، الـ Recoverable Amount بيبقى Fair Value less Costs to Sell فقط لأن صعب نحسب value in use لأصل بعمر غير محدود

الحالة الأولى — Fair Value less Costs to Sell = $2,500,000

- Carrying Value = $2,000,000
- Recoverable Amount = $2,500,000
- الـ Carrying Value أقل من الـ Recoverable Amount إذن مفيش **impairment loss**

No entry required.

Fair Value less Costs to Sell = $1,500,000 — الحالة الثانية

- Carrying Value = $2,000,000
- Recoverable Amount = $1,500,000
- الـ Carrying Value أكبر من الـ Recoverable Amount إذن فيه **impairment loss!**

Impairment Loss = 2,000,000 – 1,500,000 = $500,000

Accounts	Dr.	Cr.
Impairment loss — Broadcast license	500,000	
Broadcast license		500,000

Book value after impairment = 2,000,000 – 500,000 = **$1,500,000**

Example 2 — Patent with Impairment and Reversal (Amoc Inc.)

Amoc Inc. has a patent with carrying value of $5,000,000 at end of 2020. Value-in-use = $2,000,000, fair value less costs to sell = $1,500,000. Remaining useful life as of 1/1/2021 = 5 years. On 1/1/2023, recoverable amount = $1,800,000. **Required:** Impairment entry 31/12/2020 — Amortization 31/12/2021 — Impairment test 1/1/2023 — Amortization 31/12/2023.

الخطوة الأولى: Impairment Test في 31/12/2020

الـ Recoverable Amount = الأعلى بين Value in Use والـ Fair Value less Costs to Sell:

- Value in Use = $2,000,000
- Fair Value less Costs to Sell = $1,500,000
- الأعلى = **$2,000,000** هو الـ Recoverable Amount

الـ Carrying Value = $5,000,000 أكبر من الـ Recoverable Amount = $2,000,000 إذن فيه **impairment!**

Impairment Loss = 5,000,000 – 2,000,000 = $3,000,000

Accounts	Dr.	Cr.
Impairment loss — Patent	3,000,000	
Patent		3,000,000

Book value بعد الـ impairment **$2,000,000** = 3,000,000 – 5,000,000 = 1/1/2021 في

الخطوة الثانية: Amortization في 31/12/2021

بعد الـ impairment، الـ carrying value بقت 2,000,000$ والـ remaining life = 5 سنين.

New Amortization = 2,000,000 ÷ 5 = $400,000 per year

Accounts	Dr.	Cr.
Amortization expense — Patent	400,000	
Patent (or Accumulated amortization)		400,000

الخطوة الثالثة: Impairment Test في 1/1/2023

حساب الـ carrying value في 1/1/2023:

- Book value 2,000,000$ = 1/1/2021 في
- ناقص amortization 2021 = $400,000
- ناقص amortization 2022 = $400,000
- = **$1,200,000**

الـ Recoverable Amount الجديد = 1,800,000$ الـ Carrying Value $1,200,000 أقل من الـ Recoverable Amount $1,800,000 إذن مفيش **impairment loss** جديدة.

بس لأن فيه impairment سابقة (3,000,000$)، الفرق (600,000$) بيتعمل **reversal**.

Reversal = Min(3,000,000 خسارة سابقة، 600,000 فرق) = $600,000

Accounts	Dr.	Cr.
Patent	600,000	
Reversal of impairment loss — Patent		600,000

Book value بعد الـ reversal = 1,200,000 + 600,000 = **$1,800,000**

الخطوة الرابعة: Amortization في 31/12/2023

من 1/1/2023 لـ 31/12/2025 = 3 سنين remaining.

New Amortization = 1,800,000 ÷ 3 = $600,000 per year

Accounts	Dr.	Cr.
Amortization expense — Patent	600,000	
Patent (or Accumulated amortization)		600,000

Example 3 — Patent with Impairment and Subsequent Reversal (Alex Company)

Alex company created a patent internally and registered it on January 1, 2010 at $200,000. Useful life = 10 years. On January 1, 2011, the company successfully defended its patent paying $90,000. At end of 2011, fair value = $180,000 and value in use = $175,000. **Required:** Legal cost 1/1/2010 — Amortization 31/12/2010 — Legal fees 1/1/2011 — Amortization 31/12/2011 — Impairment 31/12/2011 — Amortization 31/12/2012.

الخطوة الأولى: Legal Cost في 1/1/2010

Accounts	Dr.	Cr.
Patent	200,000	
Cash		200,000

الخطوة الثانية: Amortization في 31/12/2010

Amortization = (200,000 − 0) ÷ 10 = $20,000 per year

Accounts	Dr.	Cr.
Amortization expense — Patent	20,000	
Patent (or Accumulated amortization)		20,000

الخطوة الثالثة: Legal Fees في 1/1/2011

الشركة نجحت في الـ defense إذن الـ $90,000 بتتكبل.

Accounts	Dr.	Cr.
Patent	90,000	
Cash		90,000

Book value 1/1/2011 على:

- Patent cost = 200,000
- ناقص (20,000) = amortization 2010
- زائد legal fees = 90,000
- **Book value 270,000$ = 1/1/2011 على**

> الخطوة الرابعة: Amortization في 31/12/2011

Remaining life 1/1/2011 على = Useful life 10 سنين ناقص سنة واحدة انقضت = **9 سنين** ⇒

Amortization = 270,000 ÷ 9 = $30,000

Accounts	Dr.	Cr.
Amortization expense — Patent	30,000	
Patent (or Accumulated amortization)		30,000

> الخطوة الخامسة: Impairment Test في 31/12/2011

Book value **240,000$** = 30,000 – 270,000 = 31/12/2011 على

الـ Recoverable Amount = الأعلى بين:

- Value in Use = $175,000
- Fair Value less Costs to Sell = $180,000
- **الأعلى = 180,000$**

الـ Carrying Value $240,000 أكبر من الـ Recoverable Amount $180,000 إذن **فيه impairment!**

Impairment Loss = 240,000 – 180,000 = $60,000

Accounts	Dr.	Cr.
Impairment loss — Patent	60,000	
Patent		60,000

Book value بعد الـ impairment **180,000$** = 60,000 – 240,000 = 1/1/2012 في

الخطوة السادسة: Amortization في 31/12/2012

Remaining life على 1/1/2012 = من 1/1/2012 لـ 31/12/2019 = **8 سنين**

Amortization = 180,000 ÷ 8 = $22,500

Accounts	Dr.	Cr.
Amortization expense — Patent	22,500	
Patent (or Accumulated amortization)		22,500

Lecture 9 — Goodwill

Example 1 — Basic Goodwill (Company A purchases Company B)

On October 1, 2023, company A purchases company B at $700,000 cash. B's balance sheet shows: Assets = $620,000, Liabilities = $200,000, Owners' equities = $420,000. Fair value of acquired assets = $800,000. **Required:** Determine and record the goodwill acquired by company A.

الخطوة الأولى: حساب Fair Value of Net Assets

- الـ Assets بنأخدها بالـ Fair Value = $800,000
- الـ Liabilities مفيش fair value محددة إذن بناخد الـ Book Value = $200,000
- الـ Owners' Equity ($420,000) مش بنأخدهاش لأن equity accounts مش بتتنقل للشركة المشترية

Fair Value of Net Assets = 800,000 – 200,000 = $600,000

الخطوة الثانية: حساب الـ Goodwill

Goodwill = Purchase Cost – Fair Value of Net Assets = 700,000 – 600,000 = $100,000

الخطوة الثالثة: قيد الشراء

Accounts	Dr.	Cr.
Assets (at Fair Value)	800,000	
Goodwill	100,000	
Liabilities (at Book Value)		200,000
Cash (Purchase Cost)		700,000

للتحقق: Dr = 800,000 + 100,000 = 900,000 / Cr = 200,000 + 700,000 = 900,000 ✓

Example 2 — Goodwill with Unrecognized Patent (Ford Inc.)

Ford Inc. purchases a parts company for $400,000 cash on January 1, 2023. The purchased company's balance sheet: Equipment $153,000 / Inventory $42,000 / Accounts Receivable $35,000 / Cash $25,000 / Accounts Payable $15,000 / Long-term liabilities $40,000 / Share capital $100,000 / Retained earnings $100,000. Fair values: Equipment = $205,000, Inventory = $122,000. Ford also detected an unrecognized patent with fair value = $18,000. **Required:** Determine and record the goodwill.

الخطوة الأولى: تحديد Fair Value لكل Asset

Asset	Book Value	Fair Value
Equipment	153,000	205,000
Inventory	42,000	122,000
Accounts Receivable	35,000	35,000
Cash	25,000	25,000
Patent (غير مسجل)	—	18,000
Total Assets (FV)		**405,000**

بالبوك فاليو عشان معطاش فير فاليو : Total Liabilities (BV) = 15,000 + 40,000 = $55,000

Fair Value of Net Assets = 405,000 – 55,000 = $350,000

الـ Share capital والـ Retained earnings مش بنأخدهاش لأنهم owners' equity بتاع الشركة المُشتراة.

الخطوة الثانية: حساب الـ Goodwill

Goodwill = 400,000 – 350,000 = $50,000

الخطوة الثالثة: قيد الشراء

Net Cash في الـ Cr = 400,000 – 25,000 = $375,000

Accounts	Dr.	Cr.
Equipment	205,000	
Inventory	122,000	
Accounts receivable	35,000	
Patent	18,000	
Goodwill	50,000	
Accounts payable		15,000
Long-term liabilities		40,000
Cash (400,000 – 25,000)		375,000

Example 3 — Goodwill Impairment (Division A in Company X)

Information about Division A (cash-generating unit) at end of 2023: Fixed assets $800,000 / Cash $50,000 / Accounts receivables $200,000 / Goodwill $200,000 / Notes payable ($900,000). Net assets = $350,000. Recoverable amount (value-in-use) = $400,000. **Required:** Record impairment of goodwill — then assuming recoverable amount = $300,000 — then assuming recoverable amount = $125,000.

الحالة الأولى — Recoverable Amount = $400,000

- Carrying Value of CGU = Net Assets = $350,000
- Recoverable Amount = $400,000
- الـ Carrying Value أقل من الـ Recoverable Amount إذن مفيش **impairment**

No entry required.

الحالة الثانية — Recoverable Amount = $300,000

- Carrying Value = $350,000 أكبر من الـ Recoverable Amount = $300,000 إذن فيه **impairment!**
- Impairment Loss of Unit = 350,000 – 300,000 = $50,000
- الـ Goodwill = $200,000
- الـ Impairment Loss ($50,000) أقل من الـ Goodwill ($200,000) إذن كل الـ impairment loss بتتسجل كـ Goodwill Impairment

Accounts	Dr.	Cr.
Impairment loss — Goodwill	50,000	
Goodwill		50,000

Goodwill after impairment = 200,000 – 50,000 = **$150,000**

> الحالة الثالثة — Recoverable Amount = $125,000

- Impairment Loss of Unit = 350,000 – 125,000 = $225,000
- الـ Goodwill = $200,000
- الـ Impairment Loss ($225,000) أكبر من الـ Goodwill ($200,000) إذن الـ Goodwill كله بيروح بـ $200,000 فقط، والـ $25,000 الباقيين بيتوزعوا على باقي assets الـ CGU

Accounts	Dr.	Cr.
Impairment loss — Goodwill	200,000	
Goodwill		200,000

Goodwill after impairment = **zero**

> ⚠️Reversal of Goodwill Impairment is not allowed under IFRS

> Example 4 — Goodwill from Purchased Division (Company X)

Company X purchased a new division on July 31, 2021 for $3,000,000. Fair value of net assets = $2,750,000. At December 31, 2023, the division's assets and liabilities are: Long-term assets including goodwill $2,400,000 / Current assets $400,000 / Long-term liabilities $700,000 / Current liabilities $500,000. Recoverable amount = $1,550,000. **Required:** Determine goodwill — Record impairment on 31/12/2023.

> الخطوة الأولى: تحديد الـ Goodwill في 31/7/2021

Goodwill = Purchase Cost – Fair Value of Net Assets = 3,000,000 – 2,750,000 = $250,000

الخطوة الثانية: Impairment Test في 31/12/2023

Carrying Value of CGU:

- Total Assets = 2,400,000 + 400,000 = $2,800,000
- Total Liabilities = 700,000 + 500,000 = $1,200,000
- **Net Assets = 2,800,000 – 1,200,000 = $1,600,000**

Recoverable Amount = $1,550,000 الـ Carrying Value $1,600,000 أكبر من الـ Recoverable Amount $1,550,000 إذن فيه **impairment!**

Impairment Loss = 1,600,000 – 1,550,000 = $50,000 , Goodwill = $250,000 and Impairment Loss ($50,000) less than Goodwill ($250,000) إذن كل الـ impairment loss بتتسجل كـ Goodwill Impairment.

Accounts	Dr.	Cr.
Impairment loss — Goodwill	50,000	
Goodwill		50,000

Goodwill after impairment = 250,000 – 50,000 = **$200,000**

Lecture 10 — Warranty (Product Guarantee) Provisions

Example 1 — Assurance-Type Warranty (Denson Machinery Company)

Denson Machinery Company sells 100 machines for $5,000 cash each by year-end. Each machine is under warranty for one year. Denson estimates warranty cost will average $200 per unit. Actual warranty costs incurred during 2023 = $4,000. **Required:** Prepare journal entries using warranty expense approach and determine effect on financial statements.

الخطوة الأولى: تسجيل البيع

Accounts	Dr.	Cr.
Cash (100 × 5,000)	500,000	
Sales revenue		500,000

الخطوة الثانية: تسجيل الـ Estimated Warranty Expense

بنقدر التكلفة الكلية وقت البيع مباشرة لأن ده الـ Assurance-Type.

Estimated Warranty = 100 machines × $200 = $20,000

Accounts	Dr.	Cr.
Warranty expense	20,000	
Warranty liability		20,000

الخطوة الثالثة: تسجيل الإنفاق الفعلي على الـ Warranty

Accounts	Dr.	Cr.
Warranty liability	4,000	
Cash		4,000

Effect on Financial Statements:

Statement of Financial Position على 31/12/2023:

- Warranty liability (Current liabilities) = 20,000 – 4,000 = **$16,000**

Income Statement لسنة 2023:

- Warranty expense = **$20,000**

> الـ Warranty Expense في الـ Income Statement دايماً هو التقديري الكلي وقت البيع مش الفعلي.
> والـ Warranty Liability في الـ Balance Sheet هي الفرق بين التقديري والفعلي المنفق.

Example 2 — Multiple Scenarios (Samsung Inc.)

In April 2024, Samsung Inc. sold 200 cellular phones on account for $1,500 each with 18-month warranty. Samsung estimates 30% of sold devices may need warranty costing $400 per device.

Required: Journal entries using warranty expense approach and effect on 2024 financial statements under:

- Case 1 — warranty costs incurred = $18,000
- Case 2 — warranty costs incurred = $24,000
- Case 3 — no warranty costs incurred

الخطوة الأولى: تسجيل البيع (ثابت في الحالات الثلاث)

Accounts	Dr.	Cr.
A/R (200 × 1,500)	300,000	
Sales revenue		300,000

الخطوة الثانية: Estimated Warranty Expense (ثابت في الحالات الثلاث)

Estimated Warranty = 30% × 200 phones × $400 = $24,000

Accounts	Dr.	Cr.
Warranty expense	24,000	
Warranty liability		24,000

الخطوة الثالثة: الإنفاق الفعلي — بيختلف حسب كل حالة

الحالة الأولى — Actual = $18,000:

Accounts	Dr.	Cr.
Warranty liability	18,000	
Cash		18,000

- Statement of Financial Position: Warranty liability = 24,000 – 18,000 = **$6,000**
- Income Statement: Warranty expense = **$24,000**

الحالة الثانية — Actual = $24,000:

Accounts	Dr.	Cr.
Warranty liability	24,000	
Cash		24,000

- Statement of Financial Position: Warranty liability = 24,000 – 24,000 = **Zero**
- Income Statement: Warranty expense = **$24,000**

الحالة الثالثة — مفيش إنفاق فعلي:

No entry required.

- Statement of Financial Position: Warranty liability = 24,000 – 0 = **$24,000**
- Income Statement: Warranty expense = **$24,000**

> في الحالات الثلاث الـ Warranty Expense في الـ Income Statement بيفضل 24,000$ ما بيتغيرش لأنه التقديري وقت البيع.

> Example 3 — Service-Type Warranty (Hamlin Auto)

On March 1, 2024, Hamlin Auto sold 10 automobiles for $30,000 each plus $900 service-type warranty for three years. During 2024, Hamlin incurred warranty costs of $5,000 ($4,000 spare parts and $1,000 cash). Revenue recognized on straight-line basis.
Required: Journal entries using warranty sales approach and effect on 2024 financial statements.

الخطوة الأولى: تسجيل البيع والـ Warranty

- Warranty Price = 10 automobiles × $900 = $9,000
- Sales Revenue = 10 × $30,000 = $300,000
- Total Cash = 300,000 + 9,000 = $309,000

Accounts	Dr.	Cr.
Cash	309,000	
Sales revenue		300,000
Unearned warranty revenue		9,000

الخطوة الثانية: تسجيل الإنفاق الفعلي على الـ Warranty

Accounts	Dr.	Cr.
Warranty expense	5,000	
Cash		1,000
Inventory (spare parts)		4,000

الخطوة الثالثة: الاعتراف بـ Earned Warranty Revenue

الشركة بتستخدم Straight-line basis. من 1/3/2024 لـ 31/12/2024 = **10** شهور.

Earned Warranty Revenue = (9,000 ÷ 3 years) × 10/12 = 3,000 × 10/12 = $2,500

Accounts	Dr.	Cr.
Unearned warranty revenue	2,500	
Warranty revenue		2,500

Effect on Financial Statements:

Statement of Financial Position على 31/12/2024:

- Unearned warranty revenue (Current liabilities) = 9,000 – 2,500 = **$6,500**

Income Statement لسنة 2024:

- Warranty expense = $5,000
- Warranty revenue = $2,500

Example 4 — Both Approaches (Future Company)

On July 1, 2024, Future company sells 1,000 units at $2,000 per unit including warranty price of $100 per unit. Company estimates 10% of units need warranty costing $500 per unit. During 2024 company spent $20,000 for servicing the one-year warranty. Revenue recognized on straight-line basis. **Required:** Journal entries using both approaches and warranty liability balance at end of 2024.

الـ Warranty Expense Approach:

الخطوة الأولى: تسجيل البيع

Accounts	Dr.	Cr.
Cash (1,000 × 2,000)	2,000,000	
Sales revenue		2,000,000

الخطوة الثانية: Estimated Warranty

Estimated Warranty = 10% × 1,000 units × $500 = $50,000

Accounts	Dr.	Cr.
Warranty expense	50,000	
Warranty liability		50,000

الخطوة الثالثة: الإنفاق الفعلي

Accounts	Dr.	Cr.
Warranty liability	20,000	
Cash		20,000

Warranty Liability Balance = 50,000 – 20,000 = $30,000 -- Income Statement: Warranty expense = $50,000

الـ Warranty Sales Approach:

الخطوة الأولى: تسجيل البيع والـ Warranty

- Warranty Price = 1,000 × $100 = $100,000
- Sales Revenue = 1,000 × (2,000 – 100) = $1,900,000

Accounts	Dr.	Cr.
Cash (1,000 × 2,000)	2,000,000	
Sales revenue		1,900,000
Unearned warranty revenue		100,000

الخطوة الثانية: الإنفاق الفعلي

Accounts	Dr.	Cr.
Warranty expense	20,000	
Cash		20,000

الخطوة الثالثة: Earned Warranty Revenue بـ Straight-line

من 1/7/2024 لـ 31/12/2024 = **6** شهور — الـ warranty period = سنة واحدة

Earned Revenue = 100,000 × 6/12 = $50,000

Accounts	Dr.	Cr.
Unearned warranty revenue	50,000	
Warranty revenue		50,000

Unearned Warranty Revenue Balance = 100,000 − 50,000 = $50,000 Income Statement: Warranty expense = $20,000 / Warranty revenue = $50,000

لو استخدمنا Actual Cost Incurred Basis بدل Straight-line: Earned Revenue = (Actual 20,000 ÷ Estimated 50,000) × 100,000 = **$40,000**	**Accounts**	**Dr.**	**Cr.**
Unearned warranty revenue	40,000		
Warranty revenue		40,000	

Lecture 11 — Accounting for Installment Sales

Example 1 — First Year (Sony Corporation)

Sony Corporation began selling goods on the installment basis on January 1, 2023. During 2023: Installment sales = $1,000,000 / Cost of installment sales = $700,000 / Cash collections = $400,000. **Required:** Prepare all journal entries using installment-sales method and determine effect on financial statements at 31/12/2023.

الخطوة الأولى: تسجيل البيع

Accounts	Dr.	Cr.
Installment A/R	1,000,000	
Installment sales		1,000,000

الخطوة الثانية: تسجيل تكلفة البيع

Accounts	Dr.	Cr.
Cost of installment sales	700,000	
Inventory		700,000

الخطوة الثالثة: إقفال حساب المبيعات والتكلفة وتسجيل الـ Deferred Gross Profit

Gross Profit = 1,000,000 – 700,000 = $300,000 (لأننا ما حصلناش Deferred بيتسجل كـ الكاش بعد)

Gross Profit Ratio = 300,000 ÷ 1,000,000 = 30%

Accounts	Dr.	Cr.
Installment sales	1,000,000	
Cost of installment sales		700,000
Deferred gross profit		300,000

الخطوة الرابعة: تسجيل الكاش المحصّل

Accounts	Dr.	Cr.
Cash	400,000	
Installment A/R		400,000

الخطوة الخامسة: تسجيل الـ Realized Gross Profit

Realized GP = GP Ratio × Cash Collected = 30% × 400,000 = $120,000

Accounts	Dr.	Cr.
Deferred gross profit	120,000	
Realized gross profit		120,000

الخطوة السادسة: إقفال الـ Realized Gross Profit

Accounts	Dr.	Cr.
Realized gross profit	120,000	
Income summary		120,000

Effect on Financial Statements:

- Installment A/R Balance = 0 + 1,000,000 – 400,000 = **$600,000** (Current Assets)
- Deferred GP Balance = 0 + 300,000 – 120,000 = **$180,000** (Current Liabilities)
- Income Statement: Realized gross profit = **$120,000**

Example 2 — With Beginning Balances (2024)

Beginning balances at end of 2023: Installment A/R = $500,000 / Deferred gross profit = $100,000 / GP ratio 2023 = 20%. During 2024: Installment sales = $800,000 / Cost of installment sales = $600,000 / Cash collections = $300,000 from 2023 sales and $400,000 from 2024 sales. **Required:** Journal entries and effect on financial statements at 31/12/2024.

الخطوة الأولى: تسجيل البيع الجديد في 2024

Accounts	Dr.	Cr.
Installment A/R	800,000	
Installment sales		800,000

الخطوة الثانية: تكلفة البيع

Accounts	Dr.	Cr.
Cost of installment sales	600,000	
Inventory		600,000

الخطوة الثالثة: الإقفال وتسجيل الـ Deferred GP لسنة 2024

Gross Profit 2024 = 800,000 – 600,000 = $200,000 GP Ratio 2024 = 200,000 ÷ 800,000 = 25%

Accounts	Dr.	Cr.
Installment sales	800,000	
Cost of installment sales		600,000
Deferred gross profit		200,000

الخطوة الرابعة: الكاش المحصّل

Total Cash = 300,000 + 400,000 = $700,000

Accounts	Dr.	Cr.
Cash	700,000	
Installment A/R		700,000

الخطوة الخامسة: الـ Realized GP

الكاش المحصّل جاي من سنتين مختلفتين وكل سنة ليها GP ratio مختلفة!

- من مبيعات 2023: 20% × 300,000 = **60,000$**
- من مبيعات 2024: 25% × 400,000 = **100,000$**
- **Total Realized GP = $160,000**

Accounts	Dr.	Cr.
Deferred gross profit	160,000	
Realized gross profit		160,000

الخطوة السادسة: الإقفال

Accounts	Dr.	Cr.
Realized gross profit	160,000	
Income summary		160,000

Effect on Financial Statements:

- Installment A/R Balance = 500,000 + 800,000 – 700,000 = **$600,000**
- Deferred GP Balance = 100,000 + 200,000 – 160,000 = **$140,000**
- Income Statement: Realized gross profit = **$160,000**

Example 3 — Three Years Data (2020, 2021, 2022)

Data:

- 2020 — Installment sales $200,000 / Cost $150,000 / Cash collections from 2020 = $80,000
- 2021 — Installment sales $250,000 / Cost $200,000 / Cash from 2020 = $100,000 / Cash from 2021 = $100,000
- 2022 — Installment sales $300,000 / Cost $210,000 / Cash from 2020 = $20,000 / Cash from 2021 = $100,000 / Cash from 2022 = $150,000

Required: Journal entries and effect on financial statements each year.

أول حاجة: حساب الـ GP Ratio لكل سنة

Year	Sales	Cost	GP	GP Ratio
2020	200,000	150,000	50,000	25%
2021	250,000	200,000	50,000	20%
2022	300,000	210,000	90,000	30%

قيود سنة 2020:

Accounts	Dr.	Cr.
Installment A/R	200,000	
Installment sales		200,000
Accounts	Dr.	Cr.
Cost of installment sales	150,000	
Inventory		150,000
Accounts	Dr.	Cr.
Installment sales	200,000	
Cost of installment sales		150,000
Deferred gross profit		50,000
Accounts	Dr.	Cr.
Cash	80,000	
Installment A/R		80,000

Realized GP = 25% × 80,000 = $20,000

Accounts	Dr.	Cr.
Deferred gross profit	20,000	
Realized gross profit		20,000
Accounts	Dr.	Cr.
Realized gross profit	20,000	
Income summary		20,000

Effect سنة 2020:

- Installment A/R = 200,000 – 80,000 = **$120,000**
- Deferred GP = 50,000 – 20,000 = **$30,000**
- Income Statement: Realized GP = **$20,000**

قيود سنة 2021:

Accounts	**Dr.**	**Cr.**
Installment A/R	250,000	
Installment sales		250,000
Accounts	Dr.	Cr.
Cost of installment sales	200,000	
Inventory		200,000
Accounts	Dr.	Cr.
Installment sales	250,000	
Cost of installment sales		200,000
Deferred gross profit		50,000
Accounts	Dr.	Cr.
Cash	200,000	
Installment A/R		200,000

الـ Realized GP:

- من 2020 sales: 25% × 100,000 = **$25,000**
- من 2021 sales: 20% × 100,000 = **$20,000**
- **Total = $45,000**

Accounts	**Dr.**	**Cr.**
Deferred gross profit	45,000	
Realized gross profit		45,000
Accounts	Dr.	Cr.
Realized gross profit	45,000	
Income summary		45,000

Effect سنة 2021:

- Installment A/R = 120,000 + 250,000 – 200,000 = **$170,000**
- Deferred GP = 30,000 + 50,000 – 45,000 = **$35,000**
- Income Statement: Realized GP = **$45,000**

قيود سنة 2022:

Accounts	**Dr.**	**Cr.**
Installment A/R	300,000	
Installment sales		300,000
Accounts	Dr.	Cr.
Cost of installment sales	210,000	
Inventory		210,000
Accounts	Dr.	Cr.
Installment sales	300,000	
Cost of installment sales		210,000
Deferred gross profit		90,000

Total Cash = 20,000 + 100,000 + 150,000 = $270,000

Accounts	**Dr.**	**Cr.**
Cash	270,000	
Installment A/R		270,000

الـ Realized GP:

- من 2020 sales: 25% × 20,000 = **$5,000**
- من 2021 sales: 20% × 100,000 = **$20,000**
- من 2022 sales: 30% × 150,000 = **$45,000**
- **Total = $70,000**

Accounts	**Dr.**	**Cr.**
Deferred gross profit	70,000	
Realized gross profit		70,000
Accounts	Dr.	Cr.
Realized gross profit	70,000	
Income summary		70,000

Effect 2022 سنة:

- Installment A/R = 170,000 + 300,000 – 270,000 = **$200,000**
- Deferred GP = 35,000 + 90,000 – 70,000 = **$55,000**
- Income Statement: Realized GP = **$70,000**